Lecture 6. Nonparametric Estimation of Basic Quantities for Right-Censored Data (Part 2)

Pointwise Confidence Intervals and Confidence Bands

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1 Recap

$$\hat{S}(t) = \prod_{t_j \le t} \Big(1 - \frac{d_j}{Y_j}\Big), \quad \hat{V}[\hat{S}(t)] = \hat{S}(t)^2 \sum_{t_j \le t} \frac{d_j}{Y_j(Y_j - d_j)}; \qquad \tilde{H}(t) = \sum_{t_j \le t} \frac{d_j}{Y_j}, \quad \sigma_H^2(t) = \sum_{t_j \le t} \frac{d_j}{Y_j^2}.$$

Define

$$\sigma_S^2(t) = \frac{\hat{V}[\hat{S}(t)]}{\hat{S}^2(t)} = \sum_{t_j \le t} \frac{d_j}{Y_j(Y_j - d_j)},$$

so $\hat{S}(t)\sigma_S(t)$ is the Greenwood SE.

2 Pointwise Confidence Intervals for $S(t)$

Motivation. Greenwood gives a direct interval that may leave $[0, 1]$; log and log-log transforms fix this.

2.1 Linear interval

Definition 6.1: Linear CI

$$\Big[\hat{S}(t) - Z_{1-\alpha/2}\hat{S}(t)\sigma_S(t),\quad \hat{S}(t) + Z_{1-\alpha/2}\hat{S}(t)\sigma_S(t)\Big].$$

Symmetric; can fall outside $[0, 1]$.

2.2 Transformed intervals: the general recipe

Derivation: delta method for a transformation g

Let g be known, monotone and differentiable. By the delta method, For known monotone differentiable g, $\mathrm{SE}\{g(\hat{S})\} \approx |g'(\hat{S}(t))|\,\hat{S}(t)\sigma_S(t)$, so a $100(1-\alpha)\%$ CI for $g(S(t))$ is

$$g(\hat{S}(t)) \pm Z_{1-\alpha/2}\,|g'(\hat{S}(t))|\,\hat{S}(t)\sigma_S(t),$$

and for $S(t)$:

$$g^{-1}\Big\{g(\hat{S}(t)) \pm Z_{1-\alpha/2}\,|g'(\hat{S}(t))|\,\hat{S}(t)\sigma_S(t)\Big\}.$$

Why transform? Faster convergence to normality, and the CI respects $0 \le S(t) \le 1$.

2.3 Log-log transformation

Theorem 6.2: Log-log CI

$$\Big(\hat{S}(t)^{1/\theta},\ \hat{S}(t)^{\theta}\Big), \qquad \theta = \exp\Big\{\frac{Z_{1-\alpha/2}\,\sigma_S(t)}{\log \hat{S}(t)}\Big\}.$$

Derivation [Supplement]. K&M use $g(u) = \log\{-\log u\}$, with derivative $g'(u) = 1/(u \log u)$ and inverse $g^{-1}(v) = \exp\{-\exp(v)\}$, so

$$|g'(\hat{S})|\hat{S}\sigma_S = \frac{\sigma_S}{|\log \hat{S}|},$$

and the CI for $g(S)$ is $\log\{-\log \hat{S}\} \pm a$, $a = Z_{1-\alpha/2}\sigma_S/|\log \hat{S}|$. Re-transforming,

$$g^{-1}\big(\log\{-\log \hat{S}\} \pm a\big) = \exp\{\log \hat{S} \cdot e^{\pm a}\} = \hat{S}^{e^{\pm a}}.$$

With $\log \hat{S} < 0$, $\theta = e^{-a} < 1$: lower $\hat{S}^{1/\theta}$, upper $\hat{S}^{\theta}$. ■

2.4 Arcsine-square-root transformation

Theorem 6.3: Arcsine-square-root CI

$$\text{lower} = \sin^2\Big\{\max\Big[0,\ \arcsin\big(\hat{S}(t)^{1/2}\big) - 0.5Z_{1-\alpha/2}\sigma_S(t)\Big(\frac{\hat{S}(t)}{1-\hat{S}(t)}\Big)^{1/2}\Big]\Big\},$$

$$\text{upper} = \sin^2\Big\{\min\Big[\frac{\pi}{2},\ \arcsin\big(\hat{S}(t)^{1/2}\big) + 0.5Z_{1-\alpha/2}\sigma_S(t)\Big(\frac{\hat{S}(t)}{1-\hat{S}(t)}\Big)^{1/2}\Big]\Big\}.$$

Derivation [Supplement]. The transformation $g(u) = \arcsin(\sqrt{u})$ has $g'(u) = 1/\{2\sqrt{u(1-u)}\}$, so $|g'(\hat{S})|\hat{S}\sigma_S = 0.5\,\sigma_S(\hat{S}/(1-\hat{S}))^{1/2}$; max / min keep the angle in $[0, \pi/2]$; $g^{-1}(v) = \sin^2 v$. ■

2.5 Log transformation

Definition 6.4: Log CI (default of R `survfit`)

$g(S) = \log S$, $g'(u) = 1/u$, so $|g'(\hat{S})|\hat{S}\sigma_S = \sigma_S$:

$$\min\Big(\exp\big\{\log\hat{S}(t) \pm Z_{1-\alpha/2}\sigma_S(t)\big\},\ 1\Big).$$

Less recommended than log-log and arcsine.

Remark 6.5: Comments

- Log-log and arcsine CIs have better coverage than linear; for very large n all agree.
- Transformed CIs are asymmetric, suited to small samples with biased, skewed estimators.

Example 6.6: Toy example, $n = 7$ [Supplement]

T_i	1.5	2.5	1.4	6.2	2.8	5.3	4.5
δ_i	1	0	0	1	1	1	0

Ordered: 1.4^+, 1.5, 2.5^+, 2.8, 4.5^+, 5.3, 6.2.

t_j	d_j	Y_j	$\hat{S}$	σ_S	Linear	Log-log	Arcsine
1.5	1	6	0.833	0.183	(0.535, 1.000*)	(0.273, 0.975)	(0.465, 1.000)
2.8	1	4	0.625	0.342	(0.207, 1.000*)	(0.142, 0.893)	(0.213, 0.949)
5.3	1	2	0.313	0.785	(0.000*, 0.794)	(0.013, 0.734)	(0.006, 0.804)
6.2	1	1	0	—	—	—	—

*Raw linear limits (1.131, 1.043, −0.169) leave $[0, 1]$. At $t = 2.8$: $\sigma_S^2 = \frac{1}{30} + \frac{1}{12} = 0.1167$, $\theta = \exp\{1.96 \times 0.3416/\ln 0.625\} = 0.241$, log-log CI $(0.625^{1/0.241}, 0.625^{0.241}) = (0.142, 0.893)$.

3 Pointwise Confidence Intervals for $H(t)$

From $\tilde{H}(t)$ and $\sigma_H(t)$:

- **Linear**: $\tilde{H}(t) \pm Z_{1-\alpha/2}\sigma_H(t)$.
- **Log** (on H; $\mathrm{SE}(\log\tilde{H}) \approx \sigma_H/\tilde{H}$): $[\tilde{H}(t)/\phi,\ \phi\tilde{H}(t)]$, $\phi = \exp\Big[\dfrac{Z_{1-\alpha/2}\sigma_H(t)}{\tilde{H}(t)}\Big]$.
- **Arcsine-square-root** (on $S = e^{-H}$):

$$\Big[-2\ln\Big\{\sin\Big(\min\Big[\frac{\pi}{2},\ \arcsin\big(e^{-\tilde{H}(t)/2}\big) + 0.5Z_{1-\alpha/2}\sigma_H(t)\{e^{\tilde{H}(t)}-1\}^{-1/2}\Big]\Big)\Big\},$$
$$-2\ln\Big\{\sin\Big(\max\Big[0,\ \arcsin\big(e^{-\tilde{H}(t)/2}\big) - 0.5Z_{1-\alpha/2}\sigma_H(t)\{e^{\tilde{H}(t)}-1\}^{-1/2}\Big]\Big)\Big\}\Big].$$

4 Confidence Bands

4.1 Pointwise intervals vs. bands

A pointwise CI satisfies $P\{L_{t_0} \leq S(t_0) \leq U_{t_0}\} = 1 - \alpha$ for a fixed t_0. Hence a plot of pointwise intervals

- is *not* an interval for the whole curve,
- and is "too narrow" compared with a band.

Definition 6.7: Confidence band

$[L(t), U(t)]$ is a $100(1-\alpha)\%$ confidence band (CB) for $S(t)$ on $t_L \leq t \leq t_U$ if

$$P\{L(t) \leq S(t) \leq U(t) \text{ for all } t_L \leq t \leq t_U\} = 1 - \alpha,$$

with t_L, t_U within the range of observed event times.

4.2 Equal probability (EP) bands

- Define
$$a_L = \frac{n\sigma_S^2(t_L)}{1 + n\sigma_S^2(t_L)}, \qquad a_U = \frac{n\sigma_S^2(t_U)}{1 + n\sigma_S^2(t_U)}, \qquad 0 < a_L \leq a_U < 1.$$
- EP bands are **proportional to the pointwise bounds**.
- Replace $Z_{1-\alpha/2}$ in Definition 6.1 and Theorems 6.2–6.3 by $c_\alpha(a_L, a_U)$ from K&M Table C.3 (e.g. $c_{0.10}(0.10, 0.40) = 2.4721 > Z_{0.95} = 1.645$).

4.3 Hall–Wellner (HW) bands

- **Not** proportional to pointwise bounds: wider for small t, narrower for large t.
- With $k_\alpha(a_L, a_U)$ from Table C.4, replace $Z_{1-\alpha/2}\sigma_S(t)$ in Definition 6.1 and Theorems 6.2–6.3 by
$$k_\alpha(a_L, a_U)\,[1 + n\sigma_S^2(t)]\,n^{-1/2}.$$

4.4 Performance and sample-size advice

- The EP *linear* band undercovers considerably for $n < 200$.
- Log-log and arcsine EP bands cover about right for $n \geq 20$ (arcsine slightly better).
- All three HW bands do well with as few as 20 events.

EP	Sample size	HW	Sample size
Linear	≥ 200	Linear	≥ 20
Log-log	≥ 20	Log-log	≥ 20
Arcsine	≥ 20	Arcsine	≥ 20

Example 6.8: Bone marrow transplant, ALL patients (K&M Table 4.3)

For 38 ALL patients KM falls from $\hat{S}(1) = 0.9737$ ($Y = 38$) to $\hat{S}(662) = 0.3531$ ($Y = 13$), flat thereafter to 2081 (censored); $\tilde{H}(662) = 1.0152$, $\sigma_H = 0.2185$. On roughly 100–600 days (K&M Fig. 4.5) both bands are wider than the pointwise intervals, and HW is wider than EP early, narrower later.

5 Selected exercises

Motivation. The log CI for H is the log-log CI for S, the arcsine CI for H comes from that for S, and without censoring the band constants reduce to empirical-process objects.

Exercise 6.1 (Log interval for H versus log-log interval for S)

Let $\tilde{H}(t) > 0$ and $\tilde{S}(t) = e^{-\tilde{H}(t)}$.
(a) Show that mapping the log-transformed CI for $H(t)$ through $S = e^{-H}$ gives $(\tilde{S}(t)^{\phi}, \tilde{S}(t)^{1/\phi})$, which is the log-log CI of Theorem 6.2 with $\hat{S}$ replaced by $\tilde{S}$ and σ_S by σ_H.
(b) Show that $\sigma_H^2(t) \le \sigma_S^2(t)$ when all $Y_j > d_j$ ($t_j \le t$), with

$$\sigma_S^2(t) - \sigma_H^2(t) = \sum_{t_j \le t} \frac{d_j^2}{Y_j^2(Y_j - d_j)}.$$

Solution [Supplement].

(a)

Step 1. $H \mapsto e^{-H}$ is decreasing: $[\tilde{H}/\phi, \phi\tilde{H}] \mapsto [\tilde{S}^{\phi}, \tilde{S}^{1/\phi}]$.

Step 2. With $\hat{S} \to \tilde{S}$, $\sigma_S \to \sigma_H$,

$$\theta = \exp\Big\{\frac{Z_{1-\alpha/2}\sigma_H}{\log \tilde{S}}\Big\} = \exp\Big\{-\frac{Z_{1-\alpha/2}\sigma_H}{\tilde{H}}\Big\} = \frac{1}{\phi}.$$

So $(\tilde{S}^{1/\theta}, \tilde{S}^{\theta}) = (\tilde{S}^{\phi}, \tilde{S}^{1/\phi})$: both are Wald intervals on the $\log(-\log S)$ scale.

(b) Termwise,

$$\frac{d_j}{Y_j(Y_j - d_j)} - \frac{d_j}{Y_j^2} = \frac{d_j\{Y_j - (Y_j - d_j)\}}{Y_j^2(Y_j - d_j)} = \frac{d_j^2}{Y_j^2(Y_j - d_j)} \ge 0;$$

summed over $t_j \le t$. The relative difference $d_j/(Y_j - d_j)$ matters only for small risk sets (the tail). ■

Exercise 6.2 (Derivation of the arcsine-square-root interval for H)

Derive the arcsine-square-root CI for $H(t)$ stated in this lecture from Theorem 6.3.

Solution [Supplement].

Step 1 (SE on the S scale). $\mathrm{SE}(\tilde{S}) \approx \tilde{S}\,\sigma_H$ (delta method), so apply Theorem 6.3 with $\hat{S} \to \tilde{S}$, $\sigma_S \to \sigma_H$.

Step 2 (rewrite in $\tilde{H}$). $\arcsin(\tilde{S}^{1/2}) = \arcsin(e^{-\tilde{H}/2})$ and

$$\Big(\frac{\tilde{S}}{1-\tilde{S}}\Big)^{1/2} = \Big(\frac{e^{-\tilde{H}}}{1-e^{-\tilde{H}}}\Big)^{1/2} = (e^{\tilde{H}} - 1)^{-1/2}.$$

So the angle interval is $[v_-, v_+]$,

$$v_\pm = \mathrm{clip}_{[0,\pi/2]}\Big\{ \arcsin\left(e^{-\tilde{H}/2}\right) \pm 0.5 Z_{1-\alpha/2}\sigma_H \left(e^{\tilde{H}} - 1\right)^{-1/2} \Big\},$$

and the CI for S is $[\sin^2 v_-, \sin^2 v_+]$.

Step 3 (back to H). Since $H = -2\log\sin v$ decreases in $v \in (0, \pi/2]$, the CI for H is

$$[-2\log\sin v_+,\ -2\log\sin v_-],$$

the stated formula; if $v_- = 0$ the upper limit is $+\infty$. ■

Exercise 6.3 (Band constants without censoring)

Let $a(t) = n\sigma_S^2(t)/\{1 + n\sigma_S^2(t)\}$, so that $a_L = a(t_L)$ and $a_U = a(t_U)$.
(a) Show that a is nondecreasing in t with values in $[0, 1)$.
(b) Without censoring, show that $a(t) = 1 - \hat{S}(t) = \mathbb{F}_n(t)$ whenever $\hat{S}(t) > 0$.
(c) Without censoring, show that the linear HW band is $\hat{S}(t) \pm k_\alpha(a_L, a_U)\, n^{-1/2}$, a band of constant width.

Solution [Supplement].

(a) $\sigma_S^2(t)$ is nondecreasing and $x \mapsto x/(1+x)$ maps $[0,\infty)$ increasingly onto $[0,1)$.

(b) By Exercise 5.1(b), $\sigma_S^2 = (1-\hat{S})/(n\hat{S})$, so $1 + n\sigma_S^2 = 1/\hat{S}$ and

$$a(t) = \frac{(1-\hat{S})/\hat{S}}{1/\hat{S}} = 1 - \hat{S}(t) = \mathbb{F}_n(t)$$

by Exercise 5.1(a): $[a_L, a_U]$ is the range of $\mathbb{F}_n$ on $[t_L, t_U]$, the Brownian-bridge time scale.

(c) By (b), $k_\alpha[1 + n\sigma_S^2(t)]n^{-1/2} = k_\alpha n^{-1/2}/\hat{S}(t)$, so the half-width is $k_\alpha n^{-1/2}$: a Kolmogorov–Smirnov-type band. ■